

Tamweel Lite: flag risky financing applications for review within a 12% capacity

10,000

synthetic applications, 22 features

7.9%

default within 90 days

10:1

cost of a missed default vs a false alarm

12%

max share sent to manual review

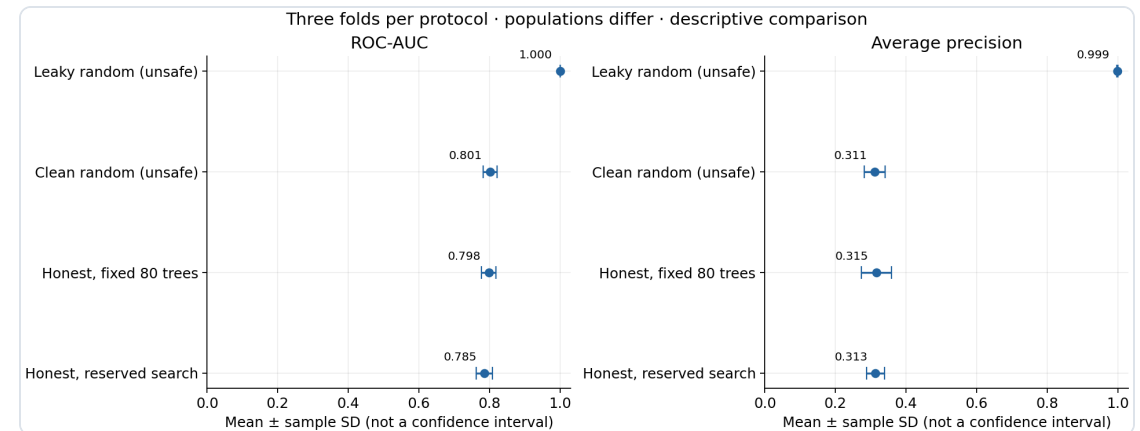
- **Decision supported:** which applications go to a simulated manual-review queue (approve vs review).
- **Target:** default within 90 days after the application, using only information known at application time.
- **Success:** lowest teaching loss $10 \times \text{FN} + \text{FP}$ while respecting 12% capacity in every period — not accuracy (flagging nobody already gives 92.4%).

Honest validation: time-forward, customer-separated, labels matured

Data → Leakage audit → Forward OOF folds → Model choice → Sigmoid calibration →

Threshold + 12% cap

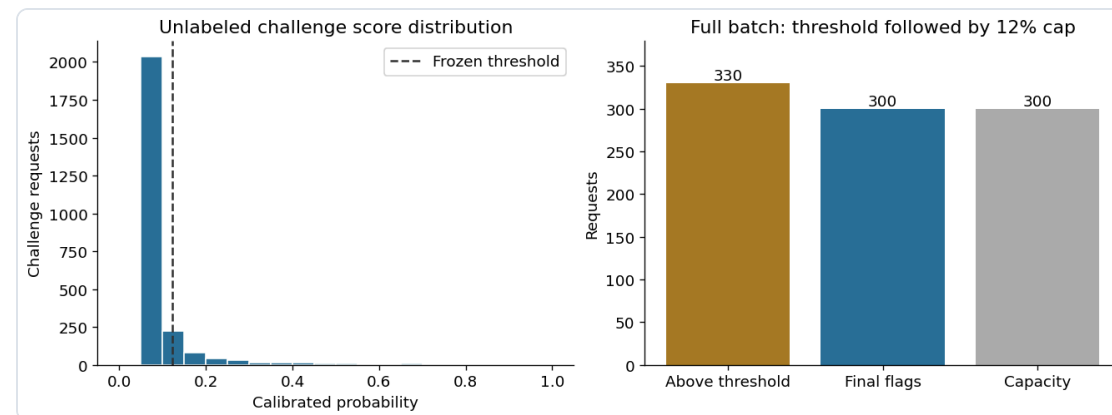
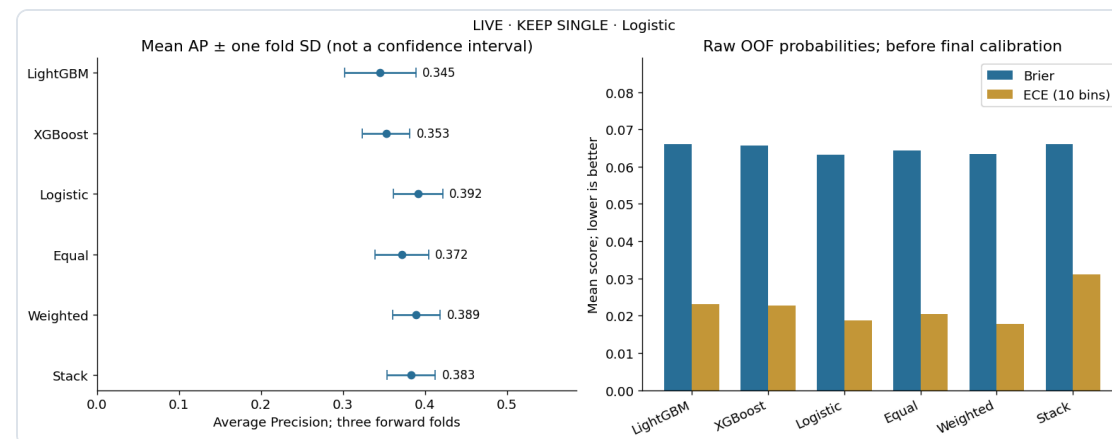
- Dropped **days_past_due_60** and **collection_calls**: known only after the application.
- Train on the past, validate on later periods; **0 shared customers**; only labels ≥ 90 days old.
- Imputation and class weights learned on training rows only; threshold chosen on OOF; calibration on a held-out period.
- Leakage would have inflated AP from 0.31 to **0.999**.



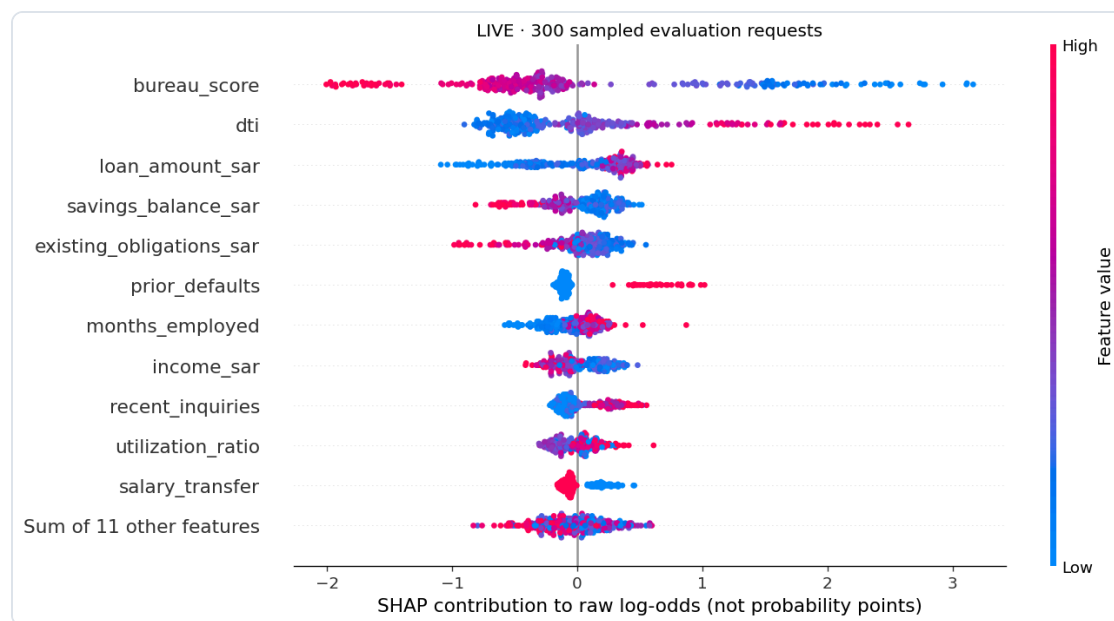
KEEP SINGLE: Logistic Regression beat boosting and every ensemble

Model	Mean AP	Fold SD	Brier
LightGBM	0.345	0.043	0.066
XGBoost	0.353	0.029	0.066
Logistic ✓	0.392	0.030	0.063
Weighted ens.	0.389	0.029	0.063
Stack	0.383	0.029	0.066

- Best ensemble lift $-0.002 < \text{one fold SD}$; model errors 0.98–0.99 correlated.
- Threshold 0.1689 on OOF: recall 46.9%, precision 34.3%, 11.4% flagged, loss 1,111.
- Challenge: 300 of 2,500 flagged (cap removed 30). No challenge accuracy claimed — labels unavailable.



bureau_score and dti drive risk — explained for the Day 4 model, not causes



- Day 4 LightGBM: permutation AP drop bureau_score 0.127, dti 0.069; SHAP in **log-odds**, not probability points.
- Example TR-009585: low bureau_score 497 (+2.27), high dti 1.28 (+1.20); raw 0.90 → calibrated 0.48.
- Day 4 calibration on a separate period: ECE 0.147 → 0.022.
- Final Logistic: sigmoid diagnostic on calibration-fit rows only (ECE 0.021 → 0.035) — not an independent test.
- **Day 4 explanations do not automatically explain the final Logistic model**; coefficients must be rechecked.

A simple, calibrated model that respects capacity — with known limits

- **Message:** honest validation showed complexity did not pay; a single Logistic model catches ~47% of defaults inside the 12% review budget.
- **Limit 1:** threshold chosen on OOF from three related periods; performance dropped in the latest period.
- **Limit 2:** regional false-positive rates 6.3%–10.3% (western highest) — descriptive, not a fairness certificate.
- **Monitor:** flag share vs 12% per batch, score drift, Brier/ECE once 90-day labels mature, regional FPR and recall.
- **Would change the decision:** a stable ensemble lift above one fold SD, or capacity overruns in new periods.
- **Repository:** github.com/shatha-alsubaie/shatha-alsubaie-advanced-machine-learning-methods-tamweel-lite